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(54) **TRANSFORMER APPARATUS AND
INTELLIGENT OIL-LEVEL DETECTION
DEVICE THEREOF**

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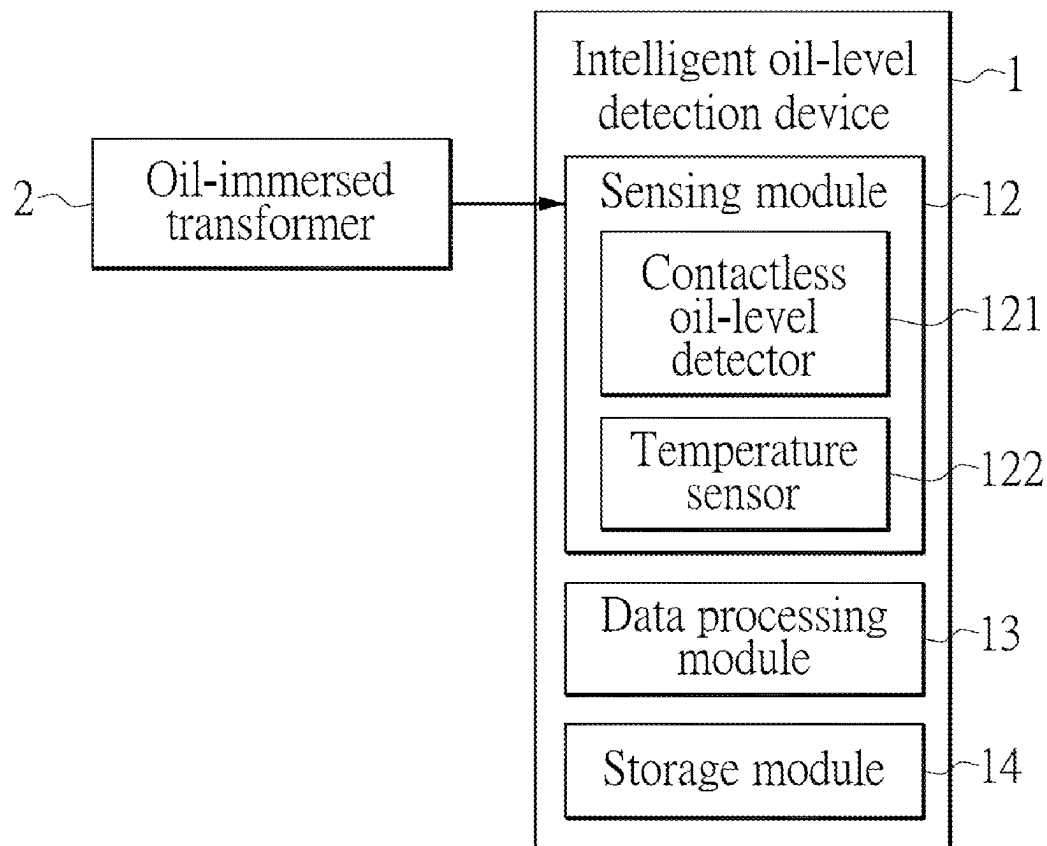
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(57) **ABSTRACT**

A transformer apparatus and an intelligent oil-level detection device are provided. The intelligent oil-level detection device for an oil-immersed transformer includes a sensing module and a data processing module. The sensing module is configured to acquire status information of an insulating oil for N consecutive days. The status information includes, at each unit time point of each day within the N days, an oil temperature, and an oil level height at the oil temperature, where N is an integer greater than 1. The data processing module is configured to generate a variation relationship curve between an oil level height and an oil temperature of each day within the N days according to the status information of the insulating oil and perform fitting processing for generating a standard reference curve.

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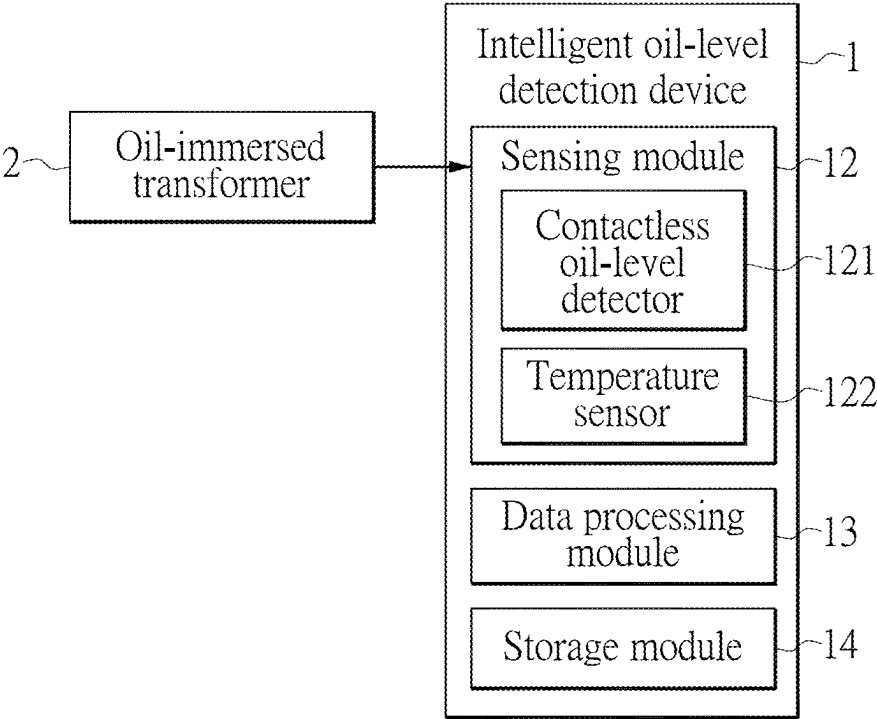
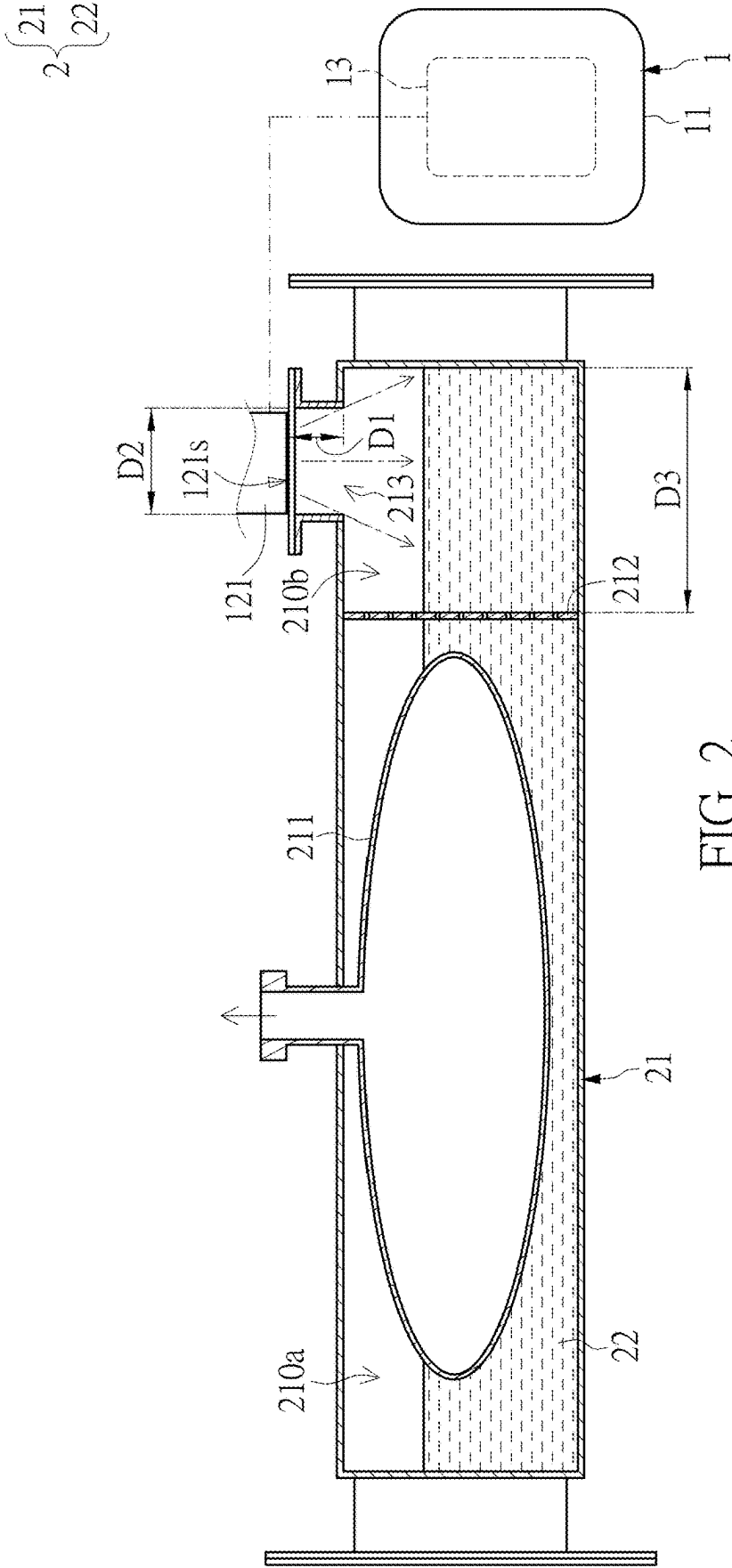


FIG. 1



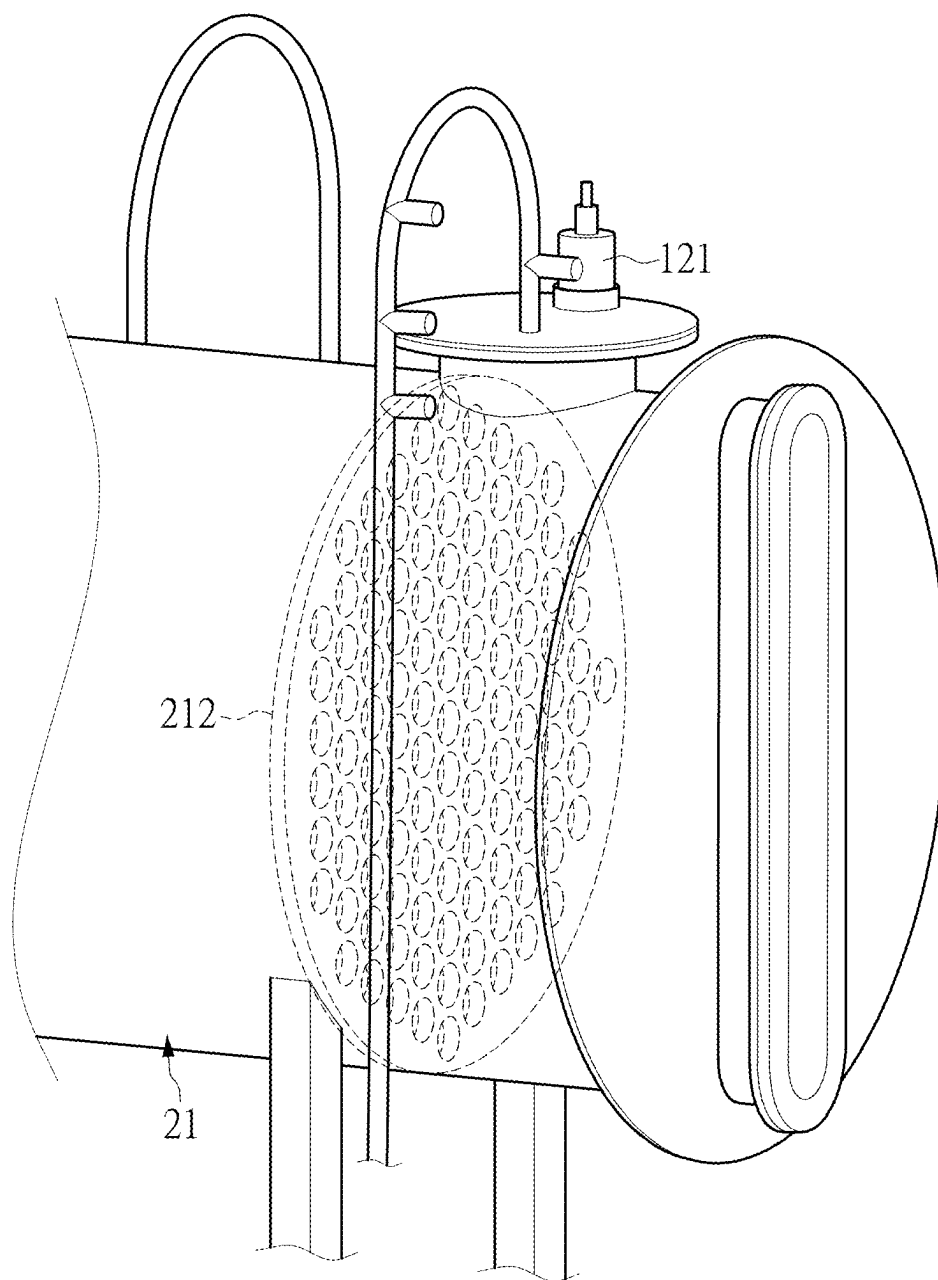


FIG. 3

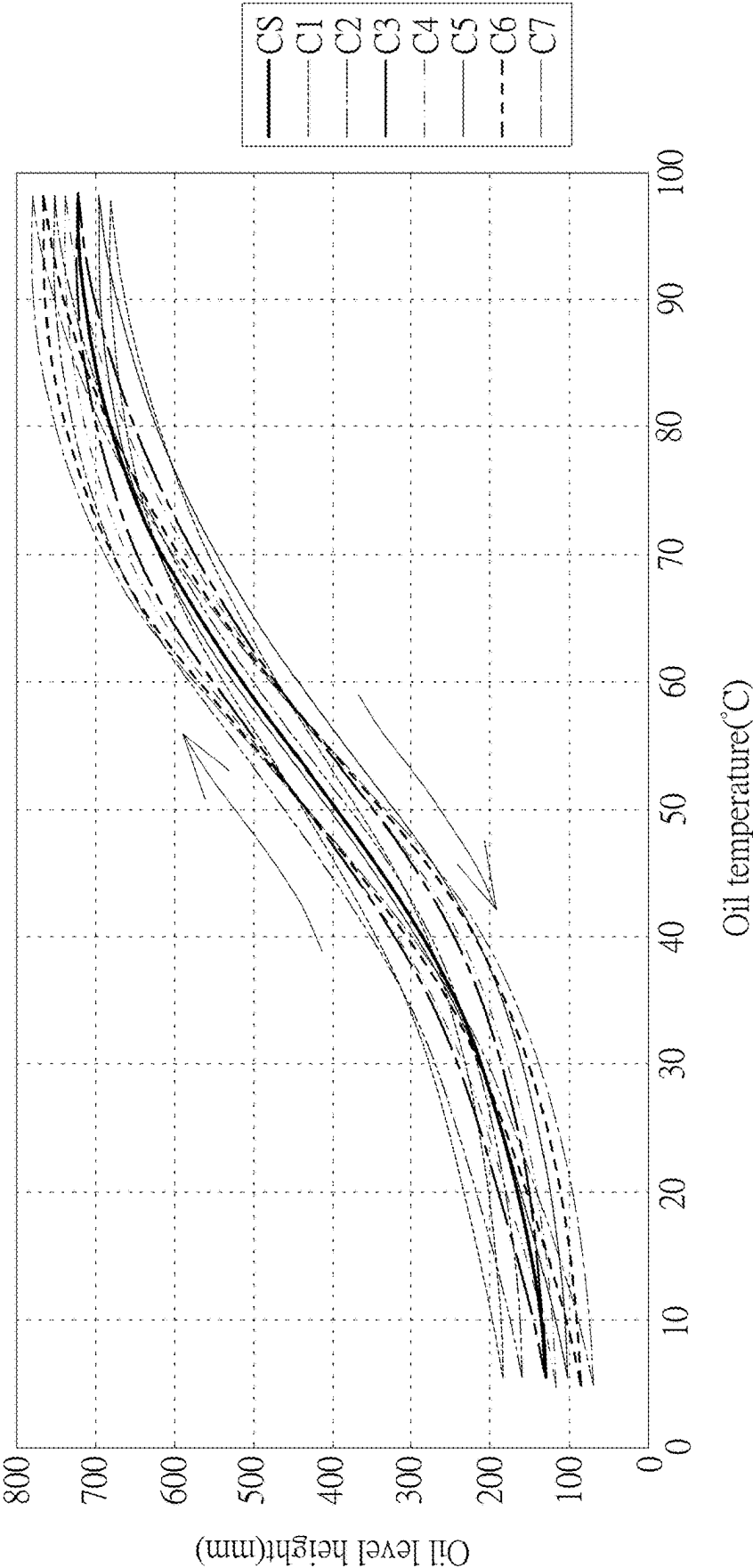


FIG. 4

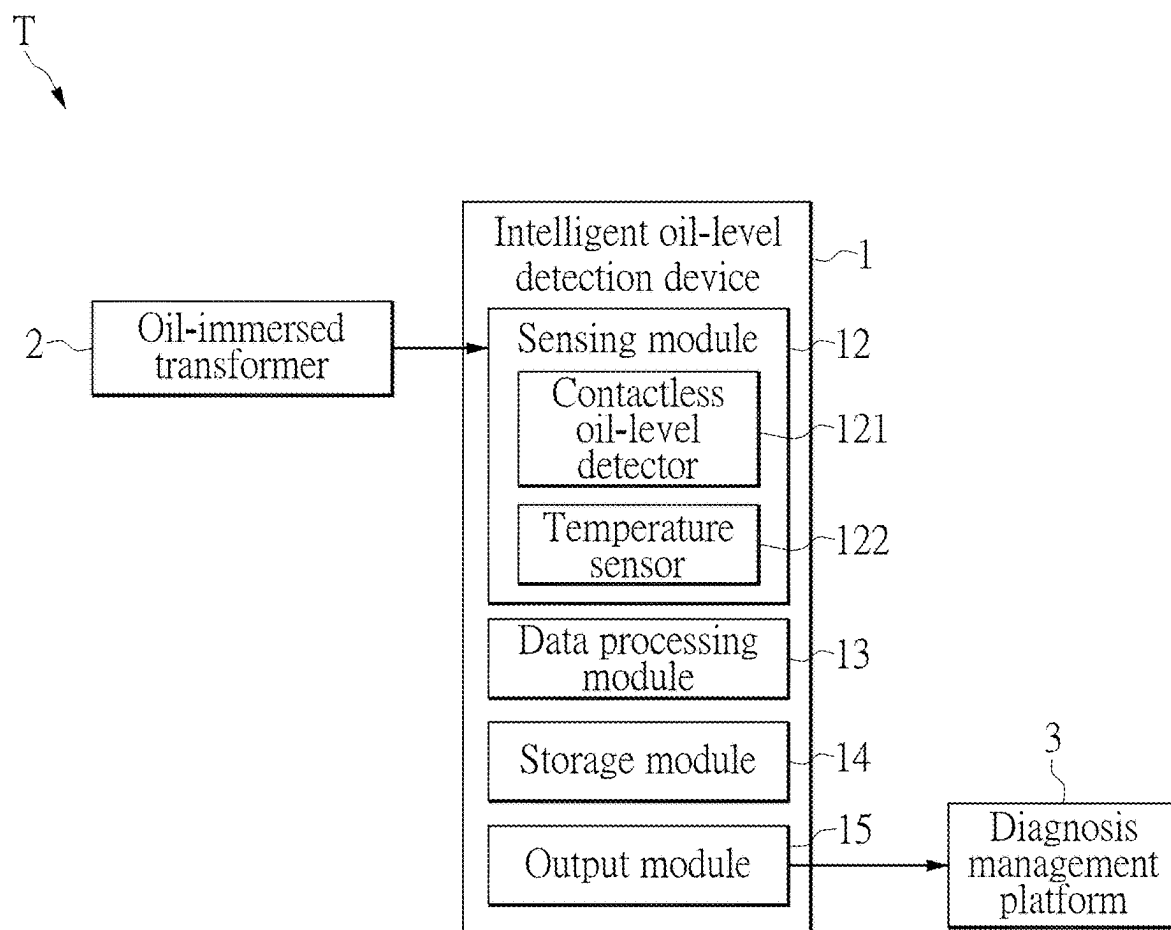


FIG. 5

TRANSFORMER APPARATUS AND INTELLIGENT OIL-LEVEL DETECTION DEVICE THEREOF

CROSS-REFERENCE TO RELATED PATENT APPLICATION

[0001] This application claims the benefit of priority to Taiwan Patent Application No. 113107804, filed on Mar. 5, 2024. The entire content of the above identified application is incorporated herein by reference.

[0002] Some references, which may include patents, patent applications and various publications, may be cited and discussed in the description of this disclosure. The citation and/or discussion of such references is provided merely to clarify the description of the present disclosure and is not an admission that any such reference is “prior art” to the disclosure described herein. All references cited and discussed in this specification are incorporated herein by reference in their entireties and to the same extent as if each reference was individually incorporated by reference.

FIELD OF THE DISCLOSURE

[0003] The present disclosure relates to power transmission and distribution apparatus, and more particularly to a transformer apparatus and an intelligent oil-level detection device.

BACKGROUND OF THE DISCLOSURE

[0004] An oil-immersed transformer is a common piece of power equipment used to decrease a high voltage to a low voltage or increase a low voltage to a high voltage, the stable operation of which is directly related to power supply reliability of an electrical grid. The main feature of the oil-immersed transformer is that an iron core and windings are immersed in an insulating oil so as to improve insulation performance and prevent failure or ignition of electrical appliances from occurring. Furthermore, the insulating oil can effectively dissipate heat, so that the oil-immersed transformer can be maintained at a relatively stable temperature to have an improved operation efficiency and lifespan.

[0005] Generally, the temperature of an insulating oil increases as the load on a transformer increases, which can cause an oil level of the insulating oil to rise due to expansion. On the contrary, the temperature of the insulating oil decreases as the load on the transformer decreases, which can cause the oil level of the insulating oil to fall due to contraction. Therefore, oil level variations can reflect the health status of the transformer. If the oil level is too high or too low, the transformer may have an abnormal condition, causing transformer failure or even an accident. For example, a pressure inside an oil tank can increase to cause oil spillage when the oil level is too high, and the insulation performance of the transformer can decrease to cause insulation breakdown when the oil level is too low.

[0006] Therefore, an oil storage device of the transformer is equipped with an oil level gauge to monitor oil level variations of the insulating oil during operation. However, the transformer may operate in different working states or environments. Even if there is a standard (such as a reference curve describing a relationship between oil level and oil temperature) that can be used to judge the correctness of the oil level variations so as to determine whether the

transformer is in normal operation, the standard cannot reflect an actual oil level height. A common procedure in the industry is to stop the operation of the transformer and then confirming an oil level once the oil level stabilizes. However, this procedure takes a long time, is complex to operate, and may cause power instability and financial losses in severe cases.

SUMMARY OF THE DISCLOSURE

[0007] In response to the above-referenced technical inadequacies, the present disclosure provides an intelligent oil-level detection device, which can provide a dedicated reference curve of correlation between oil level height and oil temperature for different transformers, so as to quickly and accurately determine whether a transformer in operation has an abnormal oil level.

[0008] In order to solve the above-mentioned problems, one of the technical aspects adopted by the present disclosure is to provide an intelligent oil-level detection device for an oil-immersed transformer that includes an oil storage device, and an insulating oil located in the oil storage device. The intelligent oil-level detection device includes a chassis, a sensing module, and a data processing module, and the chassis is configured to integrate the sensing module and the data processing module with the oil storage device. In the present disclosure, the sensing module is configured to acquire status information of the insulating oil for N consecutive days. The status information includes, at each unit time point of each day within the N days, an oil temperature, and an oil level height at the oil temperature, where N is an integer greater than 1. The data processing module is disposed in the chassis and coupled to the sensing module. The data processing module is configured to generate a variation relationship curve between oil level height and oil temperature of each day within the N days according to the status information of the insulating oil and perform fitting processing, so as to generate a standard reference curve adapted to the oil-immersed transformer. The standard reference curve is used to determine whether the oil-immersed transformer in operation has an abnormal oil level.

[0009] In order to solve the above-mentioned problems, another one of the technical aspects adopted by the present disclosure is to provide a transformer apparatus, which includes an oil-immersed transformer and an intelligent oil-level detection device. The oil-immersed transformer includes an oil storage device, and an insulating oil located in the oil storage device. The intelligent oil-level detection device includes a chassis, a sensing module, and a data processing module, and the chassis is configured to integrate the sensing module and the data processing module with the oil storage device. In the present disclosure, the sensing module is configured to acquire status information of the insulating oil for N consecutive days. The status information includes, at each unit time point of each day within the N days, an oil temperature, and an oil level height at the oil temperature, where N is an integer greater than 1. The data processing module is disposed in the chassis and coupled to the sensing module. The data processing module is configured to generate a variation relationship curve between oil level height and oil temperature of each day within the N days according to the status information of the insulating oil and perform fitting processing, so as to generate a standard reference curve adapted to the oil-immersed transformer.

The standard reference curve is used to determine whether the oil-immersed transformer in operation has an abnormal oil level.

[0010] In conclusion, the transformer apparatus and the intelligent oil-level detection device provided by the present disclosure can ascertain an abnormal condition of the oil level of different transformers as soon as possible, so as to promptly provide a notification and implement appropriate measures. Therefore, the purposes of effectively reducing troubleshooting time, preventing the transformers from being damaged, and avoiding the occurrence or expansion of the damage can be achieved.

[0011] These and other aspects of the present disclosure will become apparent from the following description of the embodiment taken in conjunction with the following drawings and their captions, although variations and modifications therein may be affected without departing from the spirit and scope of the novel concepts of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The described embodiments may be better understood by reference to the following description and the accompanying drawings, in which:

[0013] FIG. 1 is a schematic view showing an arrangement of a transformer apparatus of the present disclosure;

[0014] FIG. 2 is a schematic plan view of the transformer apparatus of the present disclosure;

[0015] FIG. 3 is a partial schematic perspective view of the transformer apparatus of the present disclosure;

[0016] FIG. 4 is a variation relationship curve between oil level height and oil temperature generated by an intelligent oil-level detection device of the present disclosure; and

[0017] FIG. 5 is a schematic view showing an arrangement of an early warning system including the transformer apparatus of the present disclosure.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

[0018] The present disclosure is more particularly described in the following examples that are intended as illustrative only since numerous modifications and variations therein will be apparent to those skilled in the art. Like numbers in the drawings indicate like components throughout the views. As used in the description herein and throughout the claims that follow, unless the context clearly dictates otherwise, the meaning of “a,” “an” and “the” includes plural reference, and the meaning of “in” includes “in” and “on.” Titles or subtitles can be used herein for the convenience of a reader, which shall have no influence on the scope of the present disclosure.

[0019] The terms used herein generally have their ordinary meanings in the art. In the case of conflict, the present document, including any definitions given herein, will prevail. The same thing can be expressed in more than one way. Alternative language and synonyms can be used for any term(s) discussed herein, and no special significance is to be placed upon whether a term is elaborated or discussed herein. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms is illustrative only, and in no way limits the scope and meaning of the present disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodi-

ments given herein. Numbering terms such as “first,” “second” or “third” can be used to describe various components, signals or the like, which are for distinguishing one component/signal from another one only, and are not intended to, nor should be construed to impose any substantive limitations on the components, signals or the like.

[0020] The normal operation of a transformer is directly related to an oil level position (i.e., an oil level height) of an insulating oil. However, when the transformer operates under a different environment, a relationship between oil level position and oil temperature may be different for the insulating oil, and it is thus difficult to determine whether an abnormal condition has occurred by using a single standard. Therefore, a technical concept provided by the present disclosure is to collect status information of each of the transformers of a power system over multiple working days, the status information includes oil level height data (i.e., measurement points) at different oil temperatures, and then to generate an optimum standard reference curve of correlation between oil level height and oil temperature by fitting processing, so as to quickly and accurately determine whether a transformer in operation has an abnormal oil level.

First Embodiment

[0021] Referring to FIG. 1 and FIG. 2, a first embodiment of the present disclosure provides an intelligent oil-level detection device 1, which is adapted to work with an oil-immersed transformer 2. The intelligent oil-level detection device 1 of the present disclosure includes a chassis 11, a sensing module 12, and a data processing module 13. The chassis 11 is configured to integrate the sensing module 12 and the data processing module 13 with an oil storage device 21 of the oil-immersed transformer 2. The data processing module 13 is disposed in the chassis 11 and coupled to the sensing module 12. In use, the sensing module 12 can acquire status information of the oil-immersed transformer 2 by detecting, for example, oil level heights and oil temperatures of an insulating oil 22. The data processing module 13 can perform programmatic steps of data analysis, numerical calculation, simulation, and correction, and the results can serve as the basis for determining abnormal conditions.

[0022] Reference is made to FIG. 4. In the present disclosure, the sensing module 12 is configured to acquire status information of the insulating oil for N consecutive days. The status information of the insulating oil includes, at each unit time point of each day within the N days, an oil temperature, and an oil level height at the oil temperature, where N is an integer greater than 1. A unit time point refers to a time point that passes one or more predetermined time intervals, which can be set arbitrarily, from a starting time point. The data processing module 13 is configured to generate variation relationship curves C1 to C7 between oil level height and oil temperature of each day within the N days according to the status information of the insulating oil and perform fitting processing. Examples of fitting processing include, but are not limited to, curve fitting. Accordingly, a standard reference curve CS adapted to the oil-immersed transformer 2 is generated to serve as the basis for determining abnormal conditions, and can be used to determine whether the oil-immersed transformer 2 in operation has an abnormal oil level. It should be noted that the standard reference curve CS as shown in FIG. 4 is only an example, and does not encompass all possible situations. In practice, the fitted curve can vary with different transformers.

[0023] In the present embodiment, the sensing module 12 can acquire current operating status parameters of the oil-immersed transformer 2 at regular time intervals (e.g., at 10-minutes or 1-hour intervals) within a predetermined period of time of each day within the N days, and the current operating status parameters mainly include oil temperatures and oil level heights of the insulating oil 22. The predetermined period of time preferably includes a high temperature period and a low temperature period of one day, which can range from, but is not limited to, 12 o'clock at night to 2 o'clock in the afternoon. The data processing module 13 can use oil level height data (i.e., measurement points) at different oil temperatures to generate a variation relationship curve between oil level height and oil temperature of the Mth day within the N days by data fitting, where M is an integer and $M \leq N$. In the example of the sensing module 12 acquiring operating status parameters of the oil-immersed transformer 2 for seven consecutive working days, the data processing module 13 can perform data fitting, so as to generate a variation relationship curve C1 between oil level height and oil temperature of the first day, a variation relationship curve C2 between oil level height and oil temperature of the second day, a variation relationship curve C3 between oil level height and oil temperature of the third day, a variation relationship curve C4 between oil level height and oil temperature of the fourth day, a variation relationship curve C5 between oil level height and oil temperature of the fifth day, a variation relationship curve C6 between oil level height and oil temperature of the sixth day, and a variation relationship curve C7 between oil level height and oil temperature of the seventh day. However, such example is not meant to limit the scope of the present disclosure.

[0024] The data processing module 13 can perform approximate simulation of fitting function curves by using an n-degree polynomial to describe a conversion function relationship between oil level height and oil temperature. Therefore, the resulting standard reference curve CS can be optimally adapted to the oil-immersed transformer 2 that operates under specific operational conditions or in a specific environment.

[0025] In practice, the sensing module 12 can include a contactless oil-level detector 121 and an oil temperature sensor 122. The contactless oil-level detector 121 can be appropriately arranged or configured to detect an oil level position (i.e., an oil level height) of the insulating oil 22 by lighting. The oil temperature sensor 122 can be appropriately arranged or configured to detect an oil temperature of a top layer of the insulating oil 22 by direct contact. The data processing module 13 can be a processing unit of a microcontroller or a general-purpose or special-purpose processor. However, the above examples are not meant to limit the scope of the present disclosure. Furthermore, the intelligent oil-level detection device 1 can further include a storage module 14 for storing data obtained by the sensing module 12 and results output by the data processing module 13. The storage module 14 can be, but is not limited to, a storage unit of the microcontroller or any form of memory.

[0026] It is worth mentioning that the data processing module 13 can repeatedly and regularly perform the above operations so as to update the standard reference curve CS dedicated to the oil-immersed transformer 2. For example, the data processing module 13 can provide a new standard reference curve CS or correct a curve feature of the standard

reference curve CS (such as a curve range or a radius of curvature or slope on the curve) every quarter or half an year according to status information of the insulating oil within another N days, where N is an integer greater than 1.

[0027] Reference is made to FIG. 5, the intelligent oil-level detection device 1 of the present disclosure can communicate with a diagnosis management platform 3 through a wired or wireless network. For example, original data or processed data can be transmitted to the diagnosis management platform 3 by an output module 15. The output module 15 can be, but is not limited to, a communication module or an output interface. Therefore, the diagnosis management platform 3 can set a threshold value according to the features of the standard reference curve CS and determine whether an actual oil level height of the insulating oil 22 is higher or lower than the threshold value. The diagnosis management platform 3 sends an early warning notification to relevant personnel such as maintenance personnel when the determination result is true. The early warning notification includes abnormal status information, hardware device information, and location information related to the oil-immersed transformer 2, such that response measures (e.g., oil replenishment) can be promptly implemented. However, the above description is disclosed for exemplary purposes only, and is not meant to limit the scope of the present disclosure. In certain embodiments, the diagnosis management platform 3 can inspect and use information in the intelligent oil-level detection device 1 when executing the application. The diagnosis management platform 3 can be implemented on a network-enabled computer device, but the present disclosure is not limited thereto.

[0028] In practice, the diagnosis management platform 3 can set an upper limit value and a lower limit value of oil level height corresponding to each temperature point within a temperature range encompassing actual oil temperatures according to the standard reference curve CS. Alternatively, the diagnosis management platform 3 can use a machine learning model to correct or reconstruct the standard reference curve CS, and can set upper and lower limit values of oil level height according to the standard reference curve CS that is corrected or reconstructed. Then, the diagnosis management platform 3 determines whether an actual oil level height is higher than the upper limit value of oil level height or is lower than the lower limit value of oil level height, and if true, determines that the oil-immersed transformer 2 in operation has an oil level abnormality. Preferably, the machine learning model can be trained with operational environment data (e.g., temperature data of surroundings), operational condition data (e.g., load data), temperature and oil level data, and historical variation relationship curves between oil level height and oil temperature.

[0029] Reference is made to FIG. 1, which is to be read in conjunction with FIG. 2 and FIG. 3. In the present disclosure, the oil storage device 21 includes an air cell 211 disposed therein to isolate the insulating oil 22 from the atmosphere, and the air cell 211 is located outside a sensing range of the contactless oil-level detector 121. The air cell 211 is configured to be in communication with the atmosphere for exhaust or suction. Furthermore, the oil storage device 21 includes a partitioning plate 212, and a first oil chamber 210a and a second oil chamber 210b in the oil storage device 21 are separate from and in communication with each other by the partitioning plate 212. More specifically, the air cell 211 is disposed in the first oil chamber

210a. The contactless oil-level detector **121** has a light emitting surface **121s** that corresponds in position to the second oil chamber **210b**. Therefore, even if the air cell **211** expands in volume, the air cell **211** will not block a light output path of the contactless oil-level detector **121**, thereby ensuring the normal operation of the contactless oil-level detector **121**.

[0030] In practice, the oil storage device **21** includes an inlet opening **213** that is arranged between the light emitting surface **121s** and the second oil chamber **210b** to allow light emitted from the light emitting surface **121s** to enter the second oil chamber **210b** and irradiate the oil surface of the insulating oil **22**. The light emitting surface **121s** and the inlet opening **213** have a predetermined distance **D1** therebetween. The inlet opening **213** has a width **D2**. The second oil chamber **210b** has an interior width **D3**. The ratio of the predetermined distance **D1**, the width **D2**, and the interior width **D3** can be 1:5 to 8:14 to 17, and preferably is 1:6:17.

Second Embodiment

[0031] Referring to FIG. 1, FIG. 3, and FIG. 4, a second embodiment of the present disclosure provides a transformer apparatus **T**, which includes an intelligent oil-level detection device **1** and an oil-immersed transformer **2**. The oil-immersed transformer **2** includes an oil storage device **21** and an insulating oil **22** located in the oil storage device **21**. The intelligent oil-level detection device **1** includes a chassis **11**, a sensing module **12**, and a data processing module **13**. The chassis **11** is configured to integrate the sensing module **12** and the data processing module **13** with the oil storage device **21**, and the data processing module **13** is disposed in the chassis **11** and coupled to the sensing module **12**. In use, the sensing module **12** is configured to acquire status information of the insulating oil for **N** consecutive days. The status information includes, at each unit time point of each day within the **N** days, an oil temperature, and an oil level height at the oil temperature, where **N** is an integer greater than 1. Furthermore, the data processing module **13** is configured to generate variation relationship curves **C1** to **C7** between oil level height and oil temperature of each day within the **N** days according to the status information of the insulating oil and perform fitting processing, so as to generate a standard reference curve **CS** adapted to the oil-immersed transformer **2**. The standard reference curve **CS** can serve as the basis for determining abnormal conditions and be used to determine whether the oil-immersed transformer **2** in operation has an oil level abnormality.

Beneficial Effects of the Embodiments

[0032] The transformer apparatus and the intelligent oil-level detection device provided by the present disclosure can ascertain an abnormal condition of the oil level of different transformers as soon as possible, so as to promptly provide a notification and implement appropriate measures. Therefore, the purposes of effectively reducing troubleshooting time, preventing the transformers from being damaged, and avoiding the occurrence or expansion of the damage can be achieved.

[0033] The foregoing description of the exemplary embodiments of the disclosure has been presented only for the purposes of illustration and description and is not intended to be exhaustive or to limit the disclosure to the

precise forms disclosed. Many modifications and variations are possible in light of the above teaching.

[0034] The embodiments were chosen and described in order to explain the principles of the disclosure and their practical application so as to enable others skilled in the art to utilize the disclosure and various embodiments and with various modifications as are suited to the particular use contemplated. Alternative embodiments will become apparent to those skilled in the art to which the present disclosure pertains without departing from its spirit and scope.

What is claimed is:

1. An intelligent oil-level detection device for an oil-immersed transformer that includes an oil storage device and an insulating oil located in the oil storage device, the intelligent oil-level detection device comprising a chassis, a sensing module, and a data processing module, wherein the chassis is configured to integrate the sensing module and the data processing module with the oil storage device;

wherein the sensing module is configured to acquire status information of the insulating oil for **N** consecutive days, and the status information includes, at each unit time point of each day within the **N** days, an oil temperature and an oil level height at the oil temperature, where **N** is an integer greater than 1;

wherein the data processing module is disposed in the chassis and coupled to the sensing module, and is configured to generate a variation relationship curve between oil level height and oil temperature of each day within the **N** days according to the status information of the insulating oil and perform fitting processing, so as to generate a standard reference curve adapted to the oil-immersed transformer;

wherein the standard reference curve is used to determine whether the oil-immersed transformer in operation has an abnormal oil level.

2. The intelligent oil-level detection device according to claim 1, wherein the data processing module is configured to modify the standard reference curve according to status information of the insulating oil within another **N** days.

3. The intelligent oil-level detection device according to claim 1, wherein the data processing module is configured to establish a conversion function relationship between oil level height and oil temperature according to the standard reference curve.

4. A transformer apparatus, comprising:

an oil-immersed transformer including an oil storage device and an insulating oil located in the oil storage device; and

an intelligent oil-level detection device including a chassis, a sensing module, and a data processing module, wherein the chassis is configured to integrate the sensing module and the data processing module with the oil storage device;

wherein the sensing module is configured to acquire status information of the insulating oil for **N** consecutive days, and the status information includes, at each unit time point of each day within the **N** days, an oil temperature and an oil level height at the oil temperature, where **N** is an integer greater than 1;

wherein the data processing module is disposed in the chassis and coupled to the sensing module, and is configured to generate a variation relationship curve between oil level height and oil temperature of each day within the **N** days according to the status infor-

mation of the insulating oil and perform fitting processing, so as to generate a standard reference curve adapted to the oil-immersed transformer;

wherein the standard reference curve is used to determine whether the oil-immersed transformer in operation has an abnormal oil level.

5. The transformer apparatus according to claim 4, wherein the sensing module includes a contactless oil-level detector that is disposed towards an oil surface of the insulating oil; wherein the oil storage device includes an air cell disposed therein to isolate the insulating oil from the atmosphere, and the air cell is located outside a sensing range of the contactless oil-level detector.

6. The transformer apparatus according to claim 5, wherein the oil storage device includes a partitioning plate, and a first oil chamber and a second oil chamber in the oil storage device are separate from and in communication with each other by the partitioning plate; wherein the air cell is disposed in the first oil chamber, and the contactless oil-level detector has a light emitting surface that corresponds in position to the second oil chamber.

7. The transformer apparatus according to claim 6, wherein the oil storage device includes an inlet opening that is arranged between the light emitting surface and the second oil chamber to allow light emitted from the light emitting surface to enter the second oil chamber and irradiate the oil surface of the insulating oil.

8. The transformer apparatus according to claim 7, wherein the light emitting surface and the inlet opening have a predetermined distance therebetween, the inlet opening has a width, the second oil chamber has an interior width, and the ratio of the predetermined distance, the width, and the interior width is 1:6:17.

9. The transformer apparatus according to claim 4, wherein the data processing module is configured to modify the standard reference curve according to status information of the insulating oil within another N days.

10. The transformer apparatus according to claim 4, wherein the data processing module is configured to establish a conversion function relationship between oil level height and oil temperature according to the standard reference curve.

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